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Enhancing respiratory virus surveillance among hospitalised children: a machine learning-based predictive model

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ABSTRACT

Background Viral respiratory tract infections (vRTIs) are a leading cause of paediatric hospitalisation and healthcare utilisation. Existing syndromic surveillance tools, including the WHO Severe Acute Respiratory Infection definition, demonstrate limited diagnostic accuracy in children whose symptom profiles vary widely. This study aimed to develop a machine learning (ML) model to predict microbiologically confirmed vRTIs in hospitalised children and to evaluate performance across age groups and viral pathogens.

Methods We conducted a retrospective cross-sectional study of 2050 paediatric patients (<18 years) admitted with acute respiratory infections to two tertiary paediatric hospitals in Canada. Predictors included age, sex, hospital transfer status, chronic comorbidity status and 22 presenting symptoms. The primary outcome was microbiologically confirmed vRTI, determined by multiplex PCR or rapid antigen testing. Six ML algorithms were trained and the best-performing model, identified by area under the receiver operating characteristic curve (auROC), was tested on age subgroups, viral pathogens and sites.

Results Among 2050 patients (median (IQR) age 2.4 (0.8–5.2) years), 1831 (89.3%) tested positive, most commonly for respiratory syncytial virus (RSV) (38.7%) and enterovirus/rhinovirus (32.8%). Logistic regression with L2 regularisation demonstrated the best performance (auROC, 0.754; 95% CI 0.697 to 0.808; sensitivity, 69.2%; specificity, 69.9%), with greatest performance among children <1 year (auROC, 0.763) and RSV cases (auROC, 0.727).

Conclusions An ML-based logistic regression model using admission data accurately predicted paediatric vRTIs, outperforming traditional syndromic surveillance definitions, especially among infants <1 year. By integrating ML models into hospital electronic medical records, healthcare systems can achieve enhanced respiratory virus surveillance, faster outbreak detection, greater diagnostic efficiency and improved pandemic preparedness.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Existing syndromic surveillance tools such as the WHO Severe Acute Respiratory Infection (SARI) definition have limited diagnostic accuracy for identifying viral respiratory tract infections (vRTIs) in children due to age-dependent and varying symptom presentations.

WHAT THIS STUDY ADDS

⇒ In this multicentre cohort of 2050 hospitalised children, machine learning (ML)-based logistic regression model leveraging routinely collected admission data accurately predicted microbiologically confirmed vRTIs, achieving an area under the receiver operating characteristic curve of 0.754, sensitivity of 69.2% and specificity of 69.9%.

⇒ This study demonstrates that interpretable, data-driven prediction models can outperform the traditional WHO SARI case definition, especially among infants <1 year, and in respiratory syncytial virus infections.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ These findings demonstrate that ML models can leverage available clinical data to support syndromic surveillance of vRTIs among paediatric patients.

⇒ By integrating ML models into hospital electronic medical records, healthcare systems can achieve faster outbreak detection, informed resource allocation, improved diagnostic efficiency and enhanced pandemic preparedness.

INTRODUCTION

Viral respiratory tract infections (vRTIs) are a leading cause of illness and mortality in children and adults worldwide. Collectively, vRTIs are responsible for substantial hospital resource utilisation and place a significant burden on healthcare systems.[1–3] Epidemics and pandemics, which can be associated with novel viral pathogens such as SARS-CoV-2 or new variants of seasonal viruses such as influenza, can change the typical seasonal patterns of viral transmission. This can result in atypical surges in respiratory illnesses, altering the proportion of vRTI-related hospitalisations among young children.[1–3] Accurate

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and timely identification of vRTIs is critical to inform infection prevention measures, initiate appropriate testing and prioritise care.[4] Early and precise detection is also essential for identifying emerging pathogens and strengthening pandemic preparedness.[5]

Hospital-based syndromic surveillance of vRTIs for detection of emerging viral respiratory outbreaks is often conducted using the case definition of severe acute respiratory infection (SARI), developed by the WHO in 2011[6] in response to the H1N1 influenza pandemic. The WHO SARI definition includes fever (self-reported or measured ≥38°C) and cough, onset in the past 10 days, and requiring hospitalisation.[6] While the WHO SARI case definition was intended to be used for viral surveillance for all age groups, its diagnostic performance among children is limited compared with adults, especially in infants <1 year.[7–9] Prior studies have described variable sensitivity and specificity across different age groups, geographic regions and virus types. A recent systematic review and meta-analysis of the performance of WHO SARI definition for influenza, based on seven studies in children, reported a sensitivity of 76% (95% CI 65% to 84%) and specificity of 31% (95% CI 20% to 44%)[7] compared with sensitivities ranging between 79% and 92% and specificities between 18% and 33% in adults.[10 11] Moreover, outbreaks may plausibly emerge from different respiratory virus families, though few studies have assessed performance for viruses beyond influenza, respiratory syncytial virus (RSV) and SARS-CoV-2 in paediatric patients.[7]

Machine learning (ML) methods offer a promising alternative to leverage routine clinical information to estimate the likelihood of a viral infection and identify emerging outbreaks in real time. It can integrate diverse clinical and sociodemographic features, capture complex interactions and improve its performance over time as more data becomes available. Therefore, we aimed to develop and evaluate a novel ML model to predict microbiologically confirmed vRTIs among hospitalised children using demographic and clinical features available at the time of hospital admission, prior to microbiologic testing.

METHODS

Study design

This study is a secondary prognostic/diagnostic cross-sectional analysis using data from the Clinical ChaRacteristics and outcomEs of hospitAlized chilDren with Acute resPiratory infecTions (READAPT)-Kids study, which was a retrospective cohort study among children aged <18 years hospitalised with acute respiratory tract infections (ARIs) from 1 July 2022 to 30 June 2023. The READAPT-Kids study was conducted at two tertiary care paediatric hospitals in Canada; the Hospital for Sick Children (SickKids) in Toronto, Ontario, and British Columbia Children's Hospital (BCCH) in Vancouver, British Columbia. Both hospitals receive high-acuity paediatric cases from surrounding community hospitals and actively participate in national respiratory viral surveillance programmes, making them highly appropriate settings for research aimed at improving respiratory virus surveillance.

Data on demographics, comorbidities, clinical symptoms and microbiological testing were extracted from electronic medical records (EMRs) by trained research assistants into a Research Electronic Data Capture (REDCap) database.[12 9] This study followed the Transparent Reporting of a Multivariable Prediction Model for Individual Prognosis or Diagnosis reporting guideline.[13]

Patient and public involvement

A patient partner was involved in the conceptualisation and design of the READAPT-Kids study, was involved in the writing and revision of this manuscript, and is a coauthor (FB). The patient partner shared their lived experience as a parent of a child with medical complexity to inform key outcomes and to assist in interpretation of the study results.

Study population

For the READAPT-Kids study, eligible patients were identified using diagnostic codes from the International Classification of Diseases, 10th Revision, Canada (online supplemental table 1). Included patients in the READAPT-Kids study were those aged <18 years at the time of hospitalisation, admitted for an acute respiratory illness, and with a clinical diagnosis caused by a suspected or confirmed viral or bacterial pathogen. Patients were excluded if they had a hospital-acquired infection, or their admission was elective, post-operative, or due to trauma.

As a secondary study of READAPT-Kids, this analysis includes patients eligible for READAPT-Kids who received multiplex PCR testing or received a positive result from a rapid antigen test (RAT). Those who only had a negative RAT result were excluded as their viral infection status could not be conclusively determined (online supplemental figure 1).

Study outcome and predictors

The outcome of interest was a microbiologically confirmed vRTI, defined as a positive result from either a multiplex PCR test or RAT administered at one of the study sites or at a community hospital prior to transfer. Testing practices varied slightly between sites, such that BCCH tested most patients who presented to their hospital, whereas SickKids was more selective. For patients transferred from a referring community hospital, repeat testing was only performed at the discretion of the clinical care provider at each respective site. RAT detected SARS-CoV-2, RSV and influenza (type A or B) only, whereas PCR testing detected these viruses and parainfluenza (PIV) (types 1–4), human metapneumovirus (hMPV), adenovirus (AdV), enterovirus/rhinovirus (EV/RV), bocavirus (BoV) and seasonal coronaviruses (sCoV) (types 229E, NL63, OC43, and HKU1). In such

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cases where a patient received both PCR testing and RAT, and where there was a notable discrepancy in the results, the PCR result was considered the 'gold standard'. Similarly, the negative class comprised patients who tested negative for all viruses on a multiplex PCR test only, as a negative result on a RAT does not imply the absence of other viruses above not detected by RAT.

Predictors for the ML models were defined a priori and included elements which would be available at the time of hospital admission or triage. The following clinically relevant predictors were identified: age, sex, transfer status (categorical; direct admission, transferred from external hospital), 24 presenting symptoms at the time of admission (online supplemental table 2), chronic comorbidity status (categorical; enrolled in complex care programme, other comorbidities than complex care, no underlying comorbidity). Of the 24 presenting symptoms, two were excluded from the model due to low heterogeneity (online supplemental table 2).

Analysis

Demographic and clinical characteristics were summarised using medians with IQRs, and counts with percentages. Data preprocessing was conducted to identify features with a proportion of missing data >10%, and collinear features with a variance inflation factor >5, neither of which occurred. Categorical data were one-hot encoded, and continuous features were scaled using the StandardScaler. A 70/30 random train-test split was performed, with the test set held out and not used for hyperparameter tuning or model training.

Five ML algorithms were trained: K-nearest neighbours, logistic regression, random forest, extreme gradient boosting (XGBoost) and support vector machine. Additionally, a voting classifier using soft voting was trained on an ensemble of the five previously mentioned algorithms. For each model, hyperparameter tuning was conducted using 10-fold grid search cross-validation (online supplemental table 3). The model was then refit with the optimal settings, and the decision threshold was determined by maximising Youden's index via 10-fold cross-validation. Model performance was evaluated on the held-out test set using the area under the receiver operating characteristic curve (auROC), area under the precision-recall curve, specificity, sensitivity, F1 score and accuracy. 95% CIs were estimated from 1000 bootstrap resamples of the test set. The best-performing model was selected based on the highest auROC score. Performance of the selected model was visualised using receiver operating characteristic curves (ROC) and calibration curves using isotonic scaling. Additional visualisation included a confusion matrix and feature importance plot. For the best performing model, random undersampling of the majority class (1:4) was performed to explore the impact of class imbalance on model performance.

Using the best performing algorithm, model performance was explored across a range of sensitivity and specificity thresholds of potential interest (75%–90%), commonly used for respiratory viral surveillance. Given clinical presentations may differ by age and viral pathogen, we further explored differences in model performance by estimating validation statistics by age subgroup (<1 year, 1 to <5 years and 5 to <18 years), pathogen (influenza, RSV, SARS-CoV-2, other viruses) and study site (BCCH and SickKids) within the test set. All analyses were conducted using Python (V.3.12.7) and the scikit-learn library (V.1.6.0).[14]

RESULTS

Description of study population

There were 2050 eligible patients hospitalised for an ARI and who had complete information on respiratory viral testing (online supplemental figure 1). There were 1238 males (60.4%) and 812 females (39.6%), and the median age at hospitalisation was 2.4 years (IQR, 0.8–5.2) (table 1). A total of 1831 patients (89.3%) had a microbiologically confirmed respiratory viral pathogen, and 219 patients (10.7%) tested negative. Among the test-positive patients, the most common pathogens were RSV (709/1831, 38.7%) and enterovirus/rhinovirus (601/1831, 32.8%). Approximately half of the patients had a chronic comorbid condition (50.8%), including 12% (245/2050) who were enrolled in the complex care programme.

The most common symptoms at time of admission were cough (82%), increased work of breathing (68.5%) and nasal congestion (60.7%) (table 1). The proportion of patients with cough (83.7% vs 67.6%), nasal congestion (62.1% vs 49.3%) and increased work of breathing (70% vs 56.2%) was higher among test-positive patients than among test-negative patients. When stratified by age groups, the prevalence of symptoms, pathogen distribution and comorbidity status varied (online supplemental table 4). RSV was more common in children <1 year (60.9%) compared with those aged 1 to <5 years (35.4%) and 5 to <18 years (16.7%). The proportion of patients with comorbidities increased with age, from 23.6% in those <1 year to 77.6% in those aged 5 to <18 years. Among test-positive patients, reported fever was less common in the <1 year group (36.6%) compared with the older groups (1 to <5 years: 54.7% and 5 to <18 years: 49.8%). Cough remained the most common symptom at presentation across all age groups, ranging from 79.5% to 82.9%.

Model performance

Of the six algorithms trained, logistic regression with ridge (L2) regularisation had the highest auROC (0.754 (95% CI 0.697 to 0.808)) and was selected as the best performing model (table 2). At the optimal threshold, this model had a sensitivity of 69.2% (95% CI 65.4% to 73.2%) and specificity of 69.9% (95% CI 58.7% to 79.8%) (table 2), and yielded comparable rates of false positives (false positive rate, 0.301) and false negatives (false negative rate, 0.308) (online supplemental table

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Table 1 Demographic and clinical characteristics by viral respiratory tract infection status*

		Viral respiratory tract infection status	
	Overall (n=2050)	**Positive (n=1831)**	**Negative (n=219)**
Age (median, IQR)	2.4 (0.8, 5.2)	2.2 (0.7, 4.8)	4.1 (1.9, 8.2)
Age groups			
1 month to <1 year	600 (29.3)	563 (30.7)	37 (16.9)
1 year to <5 years	911 (44.4)	824 (45.0)	87 (39.7)
5 years to <18 years	539 (26.3)	444 (24.2)	95 (43.4)
Sex			
Male	1238 (60.4)	1108 (60.5)	130 (59.4)
Female	812 (39.6)	723 (39.5)	89 (40.6)
Transfer status			
Transfer	759 (37.0)	688 (37.6)	71 (32.4)
Non-transfer	1291 (63.0)	1143 (62.4)	148 (67.6)
Chronic comorbid conditions			
Complex care	245 (12.0)	187 (10.2)	58 (26.5)
Other comorbidity (not complex care)	796 (38.8)	712 (38.9)	84 (38.4)
No comorbidity	1009 (49.2)	932 (50.9)	77 (35.2)
Symptoms at presentation			
Fever (reported)	993 (48.4)	878 (48.0)	115 (52.5)
Fever (measured)	584 (28.5)	518 (28.3)	66 (30.1)
Cough	1680 (82.0)	1532 (83.7)	148 (67.6)
Coryza/nasal congestion	1245 (60.7)	1137 (62.1)	108 (49.3)
Shortness of breath	359 (17.5)	315 (17.2)	44 (20.1)
Increased work of breathing	1404 (68.5)	1281 (70.0)	123 (56.2)
Fast breathing/tachypnoea	354 (17.3)	313 (17.1)	41 (18.7)
Sore throat	59 (2.9)	46 (2.5)	13 (5.9)
Apnoea	77 (3.8)	69 (3.8)	8 (3.7)
Abdominal pain	79 (3.9)	63 (3.4)	16 (7.3)
Conjunctivitis/red eyes	89 (4.3)	79 (4.3)	10 (4.6)
Cyanosis	291 (14.2)	246 (13.4)	45 (20.5)
Dehydration/reduced urine output	277 (13.5)	249 (13.6)	28 (12.8)
Diarrhoea	184 (9.0)	157 (8.6)	27 (12.3)
Fatigue/lethargy	547 (26.7)	491 (26.8)	56 (25.6)
Headache	27 (1.3)	19 (1.0)	8 (3.7)
Irritability	150 (7.3)	123 (6.7)	27 (12.3)
Myalgia/sore muscle	24 (1.2)	17 (0.9)	7 (3.2)
Poor feeding/appetite	764 (37.3)	701 (38.3)	63 (28.8)
Seizure	88 (4.3)	72 (3.9)	16 (7.3)
Wheezing	326 (15.9)	300 (16.4)	26 (11.9)
Vomiting	617 (30.1)	545 (29.8)	72 (32.9)
Rash (not hand/foot)	63 (3.1)	56 (3.1)	7 (3.2)
Other	221 (10.8)	178 (9.7)	43 (19.6)
Location			
The Hospital for Sick Children	1181 (57.6)	1024 (55.9)	157 (71.7)
British Columbia Children's Hospital	869 (42.4)	807 (44.1)	62 (28.3)
Pathogen			

Continued

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Table 1 Continued

	Overall (n=2050)	Viral respiratory tract infection status	
		Positive (n=1831)	Negative (n=219)
SARS-CoV-2	127 (6.2)	127 (6.9)	NA
Influenza	168 (8.2)	168 (9.2)	NA
Respiratory syncytial virus	709 (34.6)	709 (38.7)	NA
Parainfluenza	171 (8.3)	171 (9.3)	NA
Enterovirus/rhinovirus	601 (29.3)	601 (32.8)	NA
Human metapneumovirus	196 (9.6)	196 (10.7)	NA
Bocavirus	45 (2.2)	45 (2.5)	NA
Adenovirus	100 (4.9)	100 (5.5)	NA
Seasonal coronavirus	102 (5.0)	102 (5.6)	NA

*Data are reported as number (percentage) of participants, unless otherwise indicated.
NA, not available.

5). The calibration curve indicated that the model was underconfident, with the predicted probability underestimating the true probability of a case of vRTI occurring (online supplemental figure 2), and this did not improve with isotonic calibration. In addressing class imbalance using random undersampling, model performance did not differ substantially with an auROC of 0.744 (95% CI 0.680 to 0.802) (online supplemental table 6).

Performance of the logistic regression model varied depending on the sensitivity and specificity thresholds applied (table 3). When thresholds specifying higher sensitivity (90%) were selected, the model had a specificity of 34.2% (95% CI 24.2% to 44.9%). Conversely, when specifying specificity (90%), the sensitivity decreased to 41.7% (95% CI 37.4% to 45.9%).

Cough and nasal congestion at presentation were the strongest predictors of test-positive vRTI, while enrolment in the complex care programme, older age and irritability at presentation were the strongest predictors of test-negative vRTI (figure 1). Although a primary feature of the current SARI definitions, fever was not a strong positive or negative predictor of vRTI.

Model performance by age, virus and hospital

The performance of the logistic regression model on age subgroups varied substantially (table 4). Overall, the model performed best for patients aged <1 year (auROC, 0.763 (95% CI 0.540 to 0.940)) and worse for ages 1 to <5 years (auROC, 0.660 (95% CI 0.545 to 0.767)) and 5 to <18 years (auROC, 0.680 (95% CI 0.584 to 0.774)). The model's sensitivity decreased progressively from ages <1 year (85.5% (95% CI 80% to 90.8%)), 1 to <5 years (71.5% (95% CI 65.9% to 77.5%)) and further to ages 5 to <18 years (40.5% (95% CI 31.6% to 50%)). In contrast, specificity was highest among the older age group of 5 to <18 years at 87.2% (95% CI 75.5% to 97.1%).

Assessing specific viruses, the model performed best for detecting test-positive RSV, with an auROC of 0.727 (95% CI 0.687 to 0.767), sensitivity of 85.7% (95% CI 80.9% to 89.9%) and specificity of 47.5% (95% CI 42.4% to 52.7%) (table 4). For SARS-CoV-2, the auROC was 0.519 (95% CI 0.422 to 0.613) with a sensitivity of 60.5% (95% CI 46.3% to 74.4%) and for influenza the auROC was 0.503 (95% CI 0.431 to 0.573) with a sensitivity of 73.8% (95% CI 59.6% to 86%). The 'other pathogen' subgroup (PIV, EV/RV, hMPV, BoV, AdV and sCoV) had an auROC of 0.485 (95% CI 0.438 to 0.531) and a sensitivity of 62% (95% CI 56.3% to 67.3%). Although testing for certain viral pathogens showed improved sensitivity compared with the full cohort, the specificity was lower across all pathogen subgroups.

The model auROC was similar between SickKids and BCCH (auROC, 0.726 and 0.761, respectively), though there was higher sensitivity among patients at BCCH (79%) than at SickKids (61.2%) (online supplemental table 7).

DISCUSSION

In this study, we developed an ML model to predict microbiologically confirmed vRTIs among 2050 hospitalised children and youth using admission-level demographic and symptom data, with comparisons across age and pathogen-specific subgroups. Logistic regression with ridge (L2) regularisation demonstrated the best predictive performance, achieving an auROC of 0.754 (95% CI 0.697 to 0.808), sensitivity of 69.2% (95% CI 65.4% to 73.2%) and specificity of 69.9% (95% CI 58.7% to 79.8%). While predictive models have been used in other contexts for individual-level detection, to our knowledge, this study is novel in its application of ML to syndromic surveillance for all vRTIs, aiming to detect hospital-wide vRTI patterns for improved pandemic preparedness.

The performance of our ML model is superior compared with the performance of the WHO SARI definition in the same cohort, where sensitivity and specificity were 58% (95% CI 56% to 61%) and 49% (95%

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Table 2 Performance of machine learning models with 95% CIs

	K-nearest neighbours	Logistic regression	Random forest	Extreme gradient boosting	Support vector machine	Voting classifier
auROC	0.633 (0.558 to 0.706)	0.754 (0.697 to 0.808)	0.724 (0.666 to 0.780)	0.704 (0.641 to 0.766)	0.665 (0.604 to 0.725)	0.730 (0.672 to 0.789)
auPRC	0.935 (0.914 to 0.954)	0.950 (0.929 to 0.970)	0.947 (0.927 to 0.965)	0.940 (0.915 to 0.961)	0.934 (0.910 to 0.956)	0.948 (0.927 to 0.967)
Sensitivity	0.967 (0.952 to 0.981)	0.692 (0.654 to 0.732)	0.956 (0.938 to 0.974)	0.865 (0.837 to 0.893)	0.205 (0.170 to 0.238)	0.882 (0.855 to 0.908)
Specificity	0.110 (0.042 to 0.183)	0.699 (0.587 to 0.798)	0.219 (0.131 to 0.312)	0.356 (0.246 to 0.460)	0.973 (0.930 to 1.000)	0.384 (0.269 to 0.500)
F1	0.927 (0.912 to 0.941)	0.799 (0.771 to 0.827)	0.927 (0.912 to 0.943)	0.887 (0.866 to 0.905)	0.339 (0.291 to 0.383)	0.898 (0.879 to 0.915)
Accuracy	0.865 (0.839 to 0.891)	0.693 (0.657 to 0.730)	0.868 (0.842 to 0.894)	0.805 (0.772 to 0.834)	0.296 (0.260 to 0.330)	0.823 (0.793 to 0.850)

auPRC, area under the precision-recall curve; auROC, area under the receiver operating characteristic.

CI 42% to 55%), respectively.[9] The performance of this model further surpassed the previously proposed modified SARI definitions, which exhibited a greater trade-off between sensitivity and specificity.[7–9] Importantly, our model achieved a sensitivity of 85.5% (95% CI 80% to 90.8%) and specificity of 60% (95% CI 27.3% to 90.9%) for infants <1 year. This performance was substantially higher than sensitivities reported in previous studies using the WHO 2014 SARI case definition for this age group, ranging from 15.9%[8] to 50%,[9] with only one study[10] reporting a higher value of 74.1%. Therefore, our proposed model has the potential to improve the current syndromic surveillance strategies for children and adolescents, particularly for young children where symptom variability is high and current surveillance performance is low.

Overall, sensitivity was highest in infants <1 year and lowest in children aged 5 to <18, suggesting age-related variation in frequency of cases,[15] viral pathogens,[16] testing and hospital admission thresholds,[17] and symptoms.[18] For instance, children aged 1–6 months hospitalised with RSV are less likely to have fever but more likely to have wheezing, tachypnoea, hypoxia, and retractions, compared with older children.[18] On the other hand, specificity was highest in older patients, as they are more likely to present with specific symptoms. A recent ML study found age to be the most important feature for predicting different viral infections.[19] These findings emphasise that children cannot be treated as a homogeneous entity in prediction models and highlight the need for age-specific prediction models. While our sample size for training age-specific groups was not sufficient, future work should explore this approach to determine whether diagnostic accuracy can be further improved.

Subgroup model performance also varied by virus, where each individual pathogen performed worse than the overall model's performance, with the strongest results for RSV. Although different viruses may have different symptom presentations, such as influenza having higher rates of fever and headache, or RSV with cough, dyspnoea, and wheezing, the features are not unique to any particular pathogen.[20] Pathogens with pandemic potential may plausibly emerge from any respiratory virus family, reinforcing the importance of models trained on a diverse mix of vRTIs to maximise generalisability and improve the early identification of novel pathogens.

The study has important implications for paediatric respiratory virus surveillance and pandemic preparedness. Our model offers a data-driven, and potentially automated, approach to screen patients for vRTIs at admission, with the capacity to support proactive planning, testing prioritisation, resource allocation, and pandemic preparedness more effectively than current syndromic indicators.[21] For syndromic surveillance, high sensitivity is prioritised to ensure timely detection of cases. While WHO guidelines use a 90% confidence threshold (1.645 SD above the mean) for influenza surveillance, studies suggest optimal sensitivity thresholds

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Table 3 Logistic regression performance at varying sensitivity and specificity thresholds

	Threshold	Sensitivity	Specificity	F1	Accuracy
Sensitivity	0.90	NA	0.342 (0.242–0.449)	0.905 (0.887–0.922)	0.834 (0.805–0.862)
	0.85	NA	0.411 (0.298–0.528)	0.881 (0.861–0.901)	0.798 (0.767–0.829)
	0.80	NA	0.520 (0.406–0.645)	0.859 (0.837–0.881)	0.767 (0.735–0.802)
	0.75	NA	0.562 (0.446–0.677)	0.830 (0.805–0.855)	0.728 (0.694–0.764)
Specificity	0.90	0.417 (0.374–0.459)	NA	0.583 (0.539–0.624)	0.475 (0.432–0.515)
	0.85	0.496 (0.454–0.539)	NA	0.655 (0.616–0.692)	0.538 (0.496–0.577)
	0.80	0.614 (0.572–0.657)	NA	0.748 (0.715–0.781)	0.636 (0.598–0.675)
	0.75	0.661 (0.621–0.700)	NA	0.780 (0.749–0.809)	0.672 (0.634–0.707)

NA, not available.

of 80%–95%.[22–25] Our threshold analyses demonstrated that the model could achieve 90% sensitivity with a specificity of 34.2% (95% CI 24.2% to 44.9%), illustrating the trade-off between early detection and false positives. Sensitivity thresholds of 80% and 90% in our analysis maintained better specificity compared with other case definitions,[7] while achieving higher sensitivity. As syndromic surveillance prioritises high sensitivity, the ability of our model to achieve higher specificity at these high sensitivity levels suggests that this model could be a valuable tool for surveillance. Moreover, overtesting in high-volume settings remains a challenge, particularly in infants under 1 month who are tested more often due to lower clinical thresholds.[26] Guidelines like Choosing Wisely Canada discourage broad viral panels unless results guide management.[26] In this context, ML-based surveillance models can complement existing surveillance approaches by leveraging routinely collected clinical data to monitor respiratory virus activity and disease burden. However, there is limited evidence on how the ML models could complement public health surveillance in the real world. Therefore, future research is required

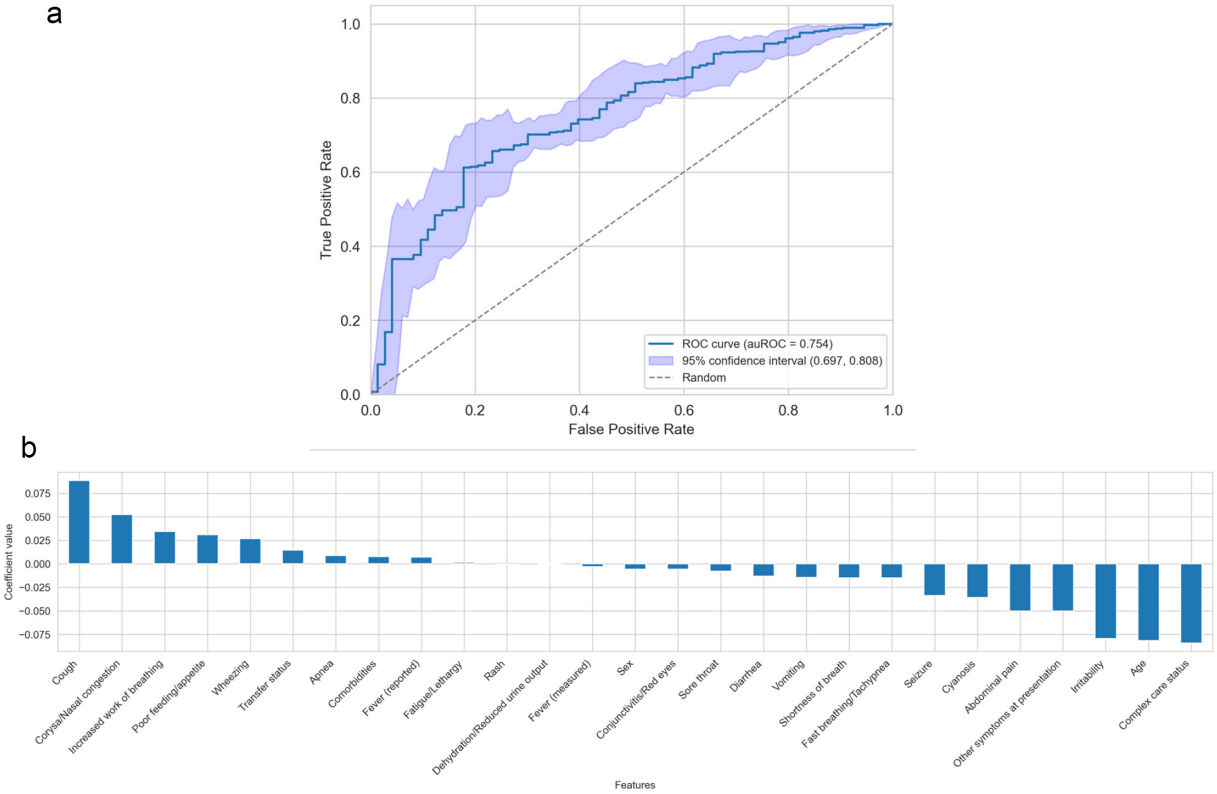


Figure 1 Receiver operating characteristic (ROC) curve and feature importance for logistic regression model. (a) ROC curve for the logistic regression model predicting viral infection. The blue line shows model performance with area under the ROC curve (auROC) of 0.754 (95% CI 0.697 to 0.808). Shaded area represents the 95% CI; dashed line represents random classification. (b) Feature importance for the logistic regression model, shown as coefficient values for individual predictors. Positive coefficients indicate features associated with increased odds of viral infection, and negative coefficients indicate features associated with decreased odds.

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Table 4 Logistic regression performance by age group and viral pathogen subgroups

	auROC	auPRC	Sensitivity	Specificity	F1	Accuracy
Age group						
Under 1 year (n=600)	0.763 (0.542–0.941)	0.965 (0.914–0.998)	0.855 (0.800–0.908)	0.600 (0.273–0.909)	0.910 (0.875–0.941)	0.841 (0.786–0.890)
1–5 years (n=911)	0.660 (0.545–0.767)	0.952 (0.919–0.977)	0.719 (0.659–0.775)	0.458 (0.259–0.667)	0.812 (0.770–0.849)	0.696 (0.641–0.751)
5–18 years (n=539)	0.680 (0.584–0.774)	0.873 (0.802–0.935)	0.405 (0.316–0.500)	0.872 (0.755–0.971)	0.560 (0.462–0.652)	0.519 (0.438–0.600)
Pathogen testing						
SARS-CoV-2+RSV+influenza (n=2050)	0.703 (0.661–0.740)	0.645 (0.589–0.703)	0.803 (0.757–0.845)	0.498 (0.444–0.549)	0.683 (0.645–0.721)	0.644 (0.607–0.680)
SARS (n=1971)	0.519 (0.422–0.613)	0.084 (0.053–0.133)	0.605 (0.463–0.744)	0.358 (0.317–0.397)	0.123 (0.081–0.169)	0.376 (0.336–0.414)
Influenza (n=2015)	0.503 (0.431–0.573)	0.063 (0.046–0.085)	0.738 (0.596–0.860)	0.363 (0.326–0.402)	0.144 (0.099–0.185)	0.389 (0.353–0.427)
RSV (n=2027)	0.727 (0.687–0.767)	0.564 (0.500–0.634)	0.857 (0.809–0.899)	0.475 (0.424–0.527)	0.620 (0.572–0.663)	0.615 (0.572–0.663)
Other* (n=1932)	0.485 (0.438–0.531)	0.539 (0.485–0.595)	0.620 (0.563–0.673)	0.348 (0.291–0.411)	0.571 (0.526–0.611)	0.496 (0.456–0.534)

*Other pathogens: parainfluenza, entero/rhinovirus, human metapneumovirus, bocavirus, adenovirus, seasonal coronavirus.
auPRC, area under the precision-recall curve; auROC, area under the receiver operating characteristic; RSV, respiratory syncytial virus.

to determine whether such approaches improve value without compromising patient safety.[27 28]

An additional strength of this approach is its potential to be used in future pandemics, where pathogen-specific microbiological testing strategies do not yet exist or are not yet widely available, offering clinicians an early-warning screening tool for existing or novel pathogens of concern that may be suggestive of emerging respiratory threats.[11] Integration of this model into hospital-based EMRs could enable automated detection of both seasonal trends and unusual outbreaks of vRTIs, enhancing efficiency and timeliness relative to manual recognition.

Implementation and governance of ML models would involve multiple stakeholders across healthcare systems. Public health authorities and Infection Prevention and Control (IPC) leaders would play crucial roles in integrating hospital-based model outputs into regional respiratory surveillance frameworks and early outbreak detection systems. IPC teams could use real-time trends to identify unusual increases in viral-like presentations, anticipate isolation room demand, guide personal protective equipment planning, and initiate enhanced outbreak surveillance when thresholds are exceeded. While hospital operational leaders may use the model to anticipate staffing needs, capacity of beds, and isolation resource allocation, frontline clinicians, such as emergency physicians and paediatric hospitalists, may use the model to contextualise admission decisions and priority setting.

This research adds to the growing body of literature on ML-based prediction models for infectious diseases.[29–31] Although clinical application studies in paediatric populations are limited, results from adult populations demonstrated promising real-life applications of predictive modelling powered by ML.[29] LASSO-based models can support real-time forecasts for endemic infections.[32 33] For instance, COVID-GRAM, a web-based calculator, can predict COVID-19 outcomes in hospitalised patients.[34]

Aligning with previous literature, our efforts suggest a promising avenue for the translation of predictive models into tools that can be integrated into hospital systems, aiding in pandemic preparedness and response, thereby potentially minimising disease burden of respiratory infections on both children and paediatric care centres.

This study has several strengths. It is one of the largest paediatric-focused ML prediction studies for vRTIs and includes data from two large tertiary care children's hospitals that are geographically separate and serve diverse populations. This model relies solely on data available at admission and does not require laboratory or imaging results, making it a clinically practical tool, especially in fast-paced hospital environments where diagnostic testing may be delayed or unavailable. The easily interpretable logistic regression demonstrated comparable performance to that of more complex algorithms like support vector machines and XGBoost, underscoring prior research that regularised logistic regression can perform as well as more complex ML methods in structured clinical data settings and healthcare modelling.[35]

Nevertheless, our study has limitations. First, this study was retrospective in design, relying on clinician-driven testing decisions, and clinical documentation and laboratory testing performed as part of routine care. This may have introduced verification bias or inconsistencies

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in documentation. The retrospective design may have also restricted the ability to assess how the model would perform prospectively in real-time clinical workflows. Moreover, data collection was conducted during the lifting of protective non-pharmaceutical interventions implemented for the COVID-19 pandemic, leading to an atypical season with surges of RSV and influenza in paediatric patients. In addition, the model was developed in a cohort consisting only of patients admitted with a suspected ARI, where 89% of tested patients were positive for at least one virus. This may not reflect a typical respiratory season and could reduce model performance in more typical seasons and broader patient populations. This class imbalance also limited the ability to train age-specific and pathogen-specific models, as there was an inadequate number of negative cases across all subgroups. Importantly, our model excluded patients who were not tested for viral pathogens, which may lead to selection bias and limit generalisability. The potential application of our model to non-viral respiratory infections warrants further investigation. Additionally, since our dataset came from two Canadian tertiary hospitals that accept transfers from community hospitals and may treat disproportionately ill patients or severe cases, this may lead to bias and limit the generalisability, thus external validation of this model in various settings across multiple seasons is required.[36] Moreover, race and ethnicity data were not systematically available in the EMRs used for this study and were therefore not included in model development. This may have limited our ability to assess performance across racial and ethnic subgroups. Future validation studies should examine model performance across diverse populations and incorporate equity-focused evaluation to ensure fair and unbiased implementation. Future research on building models that focus on paediatric age subgroups is necessary to improve model performance and clinical validity.[36–38] Lastly, the use of ML models based on routinely collected data may raise important considerations regarding data privacy, transparency and algorithmic biases for patients and families, and perceived understandability and trustability concerns for clinicians.[39 40] Real-world implementation of this model following further validation will need to adhere to privacy policies governing the use of EMR-based ML models.

In conclusion, in this multicentre cohort of hospitalised children, we developed and evaluated an ML model using admission-level clinical and demographic data to predict microbiologically confirmed vRTIs. The findings suggest that ML models based on routinely collected clinical data can support improved syndromic surveillance. By developing and validating interpretable data-driven prediction models, public health can better integrate automated symptom-based surveillance into hospital EMRs. This could enable infection control teams to better detect unusual surges in respiratory viral infections in real-time, optimise resource allocation, and provide early-warning tools for emerging pathogens in settings where accurate microbiological testing is unavailable or infeasible. This can ultimately improve pandemic preparedness, patient care and clinical outcomes for paediatric populations and the hospitals that serve them. Exploration of age-specific models, application to non-viral infections and external validation across various settings and multiple seasons can further improve model's performance and generalisability.

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